

EE2020/EE2026 (Part 1)
Tutorial 2 - Questions

Number systems

1. (a) What is the ordinary 8 bit binary equivalent of the decimal number 250?
(b) What decimal number does the above bit pattern correspond to if interpreted as
 - i) a signed magnitude number
 - ii) a 1's complement number and
 - iii) a 2's complement number?
2. Show how the following can be added in 2's complement notation using 8-bit arithmetic
 - (a) $(-1) + 45$
 - (b) $-128 + (-60)$
3. Compute and give the final answer in 2's complement notation:
 $(10100)_{2's} + (00100)_{SM}$
4. Convert the 8421 BCD number 0100011000100011 into decimal.